

Scenario:

Dr. Robert Quincy is on vacation in the Hawaiian estate of his good friend Robin Masters. During his vacation, Detective Angela Lansbury is acting director of the UNCC forensics lab. Det. Lansbury is on the case of one of North Carolina's best-known villains. The "goon for hire" Rett Haring. Rett is known to work for several different supervillains, including Malcom Monarch and Gene Poole. Occasionally, Rett has been known to engage in petty thievery to earn a living in between acting as a minion. Detective Lansbury has brought you a Chat App database that one of the lab tech's extracted with UNCC's SIS created Mobile Electronics Technical Analysis Lab. The lab tech couldn't get METAL to parse the database, so you, the expert, will be trying to make sense of who Rett has been talking to and what he's been talking about.

Forensic Acquisition & Exam Preparation:

Having booted the forensic workstation to Windows 11, I verified the database files using DB Browser for SQLite.

Following this, all request items were satisfied using DB Browser.

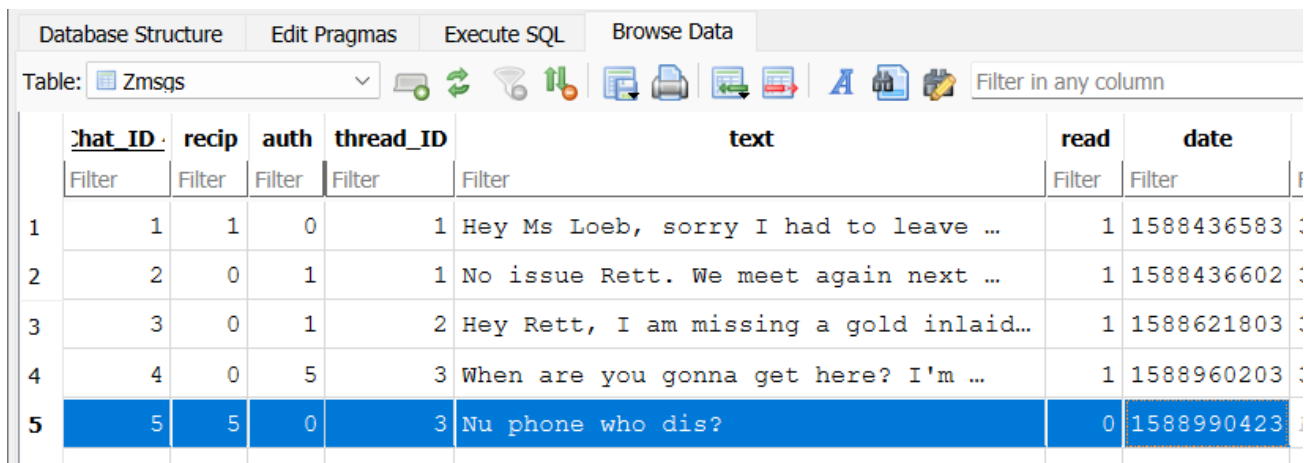
Findings and Report (Forensic Analysis)

Using the DB Browser for SQLite tool, I answered all the questions asked by the referring agency.

Each item is addressed separately below.

1) What was the date of contact for the user who replied, "Nu phone who dis".

Opening DB Browser and uploading the ChattyMcChatFace database with the SQLite tool, I was able to view the table data under the "Browse Data" tab. Under this tab, I was able to select specific tables and view them without having to create a schema. Upon doing this, I viewed the specific message "Nu phone who dis." There, I saw many details such as the Chat_ID, recipient, author, thread_ID, text, if the message was read, date, and location. The date of contact of the user who replied "Nu phone who dis" was 1588990423.



Database Structure Edit Pragas Execute SQL Browse Data						
Table: Zmsgs						
chat_ID	recip	auth	thread_ID	text	read	date
Filter	Filter	Filter	Filter	Filter	Filter	Filter
1	1	1	0	1 Hey Ms Loeb, sorry I had to leave ...	1	1588436583
2	2	0	1	1 No issue Rett. We meet again next ...	1	1588436602
3	3	0	1	2 Hey Rett, I am missing a gold inlaid...	1	1588621803
4	4	0	5	3 When are you gonna get here? I'm ...	1	1588960203
5	5	5	0	3 Nu phone who dis?	0	1588990423

Figure 1: DB Browser Browse Data of "Zmsgs" table.

2) Which users (by their email address) were involved in that conversation?

Using DB Browser, specifically the Browse Data and table views, I was able to identify the two users involved in that conversation. The recipient of the message is user ID 5, while the author's ID is 0. Using that information and cross-checking it with the zContacts table, I was able to find their information. The user ID whose number is 5 is Kamoze, email being kamoze@chattymail.com, the author ID whose number is 0 is Rett, email being rett.harring@outlook.com.

App_ID	Ver	New	Last	OS_ver	First	Alt	Avatar
1	0.9	1.1	Harring	8	Rett	rett.harring@outlook.com	BLOB

Figure 2: DB Browser zConstants Browse Data table showcasing Rett Harring's Information.

Contact_ID	Cnum	Cnick	Clast	Calt	Cavatar	Cfirst
5	9105552401	Hotstepper	Kamoze	kamoze@chattymail.com	BLOB	Ini...

Figure 3: DB Browser zContacts Browse Data table showcasing Kamoze's information.

3) What does Ms. Loeb look like?

Using DB Browser to view the ChattyMcChatfaceChats database to find what Ms. Loeb looks like, I first went through the table named Thread. Once looking at the Browse Data of the Thread table, selecting Ms. Loeb's contact information, followed by selecting "BLOB" under "Cavatar" for Ms. Loeb's name, I was able to see her photo.

Contact_ID	Cnum	Cnick	Clast	Calt	Cavatar	Cfirst
1	8285550496	OMG_glasses	Loeb	Firecracker@chattymail.com	BLOB	Lisa

Figure 4: DB Browser zContacts Browse Data table showcasing Ms. Loeb's information.

4) What is the item that appears to have been stolen and when was the message sent?

Using DB Browser under the zMessages table using the Browse Data tab to view the tables I was able to uncover what item appears to have been stolen. The item that appears to have been stolen is a "gold inlaid guitar pick"; the time the message was sent was at 1588621803. Where this information was recovered from was the message sent from Ms. Loeb to Rett.

Message_ID	Cnum	Cnick	Clast	Calt	Cavatar	Cfirst	Ctime
3	0	1	2	Hey Rett, I am missing a gold inlaid...	1	1588621803	35°14'03.5"N 80°50'30.9"W

Figure 5: DB Browser zMessages Browse Data table showcasing Ms. Loeb's message to Rett.

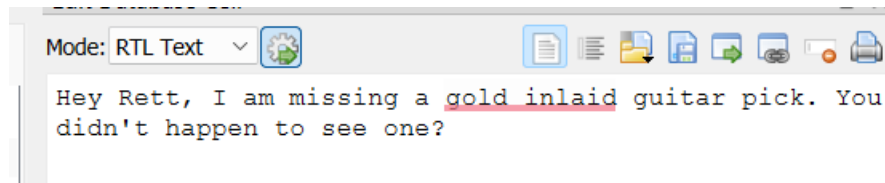


Figure 6: DB Browser Ms. Loeb's full message to Rett.

5) Who has Rett sent messages to most often (tell me their email)?

Through DB Browser, I was able to view the different messages that Rett has sent most often. Cross-checking throughout the Zmsgs and comparing how many messages were sent to the different users, I was able to see that the user ID 3 was receiving the most messages from Rett. The email corresponding to the user ID 3 is OneZeroZeroPct@chattymail.com.

11	11	3	0	6	Hey - nah I got it. You seen Gene ...	1	1589950899	NULL
12	12	4	0	7	I need to share my plan with Gene, ...	1	1589950899	NULL
13	13	4	0	7	Can't get ahold of Gene. Can you sen...	0	1589989999	NULL
14	14	4	0	7	SSBoYXZlIGZvdW5kIGEd2F5IHRvIG1ha2Ug...	0	1589989999	NULL
15	15	3	0	6	He said he "found" what I was lookin...	1	1589991234	NULL
16	16	3	0	6	Its this set of beauties	1	1589991235	NULL
17	17	3	0	6	FF D8 FF E0 00 10 4A 46 49 46 00 01 ...	1	1589991239	NULL

Figure 7: DB Browser Browse Data tab of Zmsgs table.

Table: zContacts							
	Contact_ID	Cnum	Cnick	Clast	Calt	Cavatar	Cfirst
	Filter	Filter	Filter	Filter	Filter	Filter	Filter
1	1	8285550486	OMG_glasses	Loeb	Firecracker@chattymail.com	BLOB	Lisa
2	2	9195551337	HocusPocus	vanLeer	Focus@chattymail.com	BLOB	Thijs
3	3	2525550100	GyspyWoman	Waters	OneZeroZeroPct@chattymail.com	BLOB	Crystal
4	4	9805551212	AdamAnt	Goddard	TwoShoes@chattymail.com	BLOB	Stuart
5	5	9105552401	Hotstepper	Kamoze	kamoze@chattymail.com	BLOB	Ini...
6	6	7045559991	Peppermints	Munford	Strawberry@chattymail.com	BLOB	Greg
7	7	8006492568	MixAlot	Ray	GotBack@chattymail.com	BLOB	Anthony

Figure 8: DB Browser Browse Data tab of zContacts table, highlighted is user ID 3's information.

6) Bonus: Can you tell me what the text was in the secret message?

Looking at the zMsgs table using the Browse Data tab of DB Browser, the message sent by Rett contains encrypted text. The original text is as follows:

ISSBoYXZlIGZvdW5kIGEd2F5IHRvIG1ha2UgdGhlIGRydWdzIHVuZGV0ZWN0YWJsZSBieSB0aG
UgZG9ncyBhdCB0aGUgY2hlY2twb2ludC4gSXQncyBteSBvd24gc3BIY2lhbCBtaXggby2Ygc2FnZSw
gcm9zZW1hcnksIHRoeW1lLCB2YXBvcnViIGFuZCBjYXJvbGluYSByZWFWZXIgb2lsIQ==

After using a Base64 Decoder, the message I decoded was “I have found a way to make the drugs undetectable by the dogs at the checkpoint. It’s my own special mix of sage, rosemary, thyme, vaporub and carolina reaper oils!”

Database Structure		Edit Pragma	Execute SQL	Browse Data	Edit Database Cell	
Table: Zmsqs						
chat_ID	recip	auth	thread_ID	text	read	date
Filter	Filter	Filter	Filter	Filter	Filter	Filter
7	7	1	0	2 Nope. I haven't seen it.	1	1588629811 NULL
8	8	5	0	5 Oh, Rett.. sry mis yr msg. Been out ...	1	1589140823 NULL
9	9	0	1	2 Okay, let me know if you do find it.	1	1589950821 35°14'03.5"N
10	10	0	3	6 Crys. You need Gene's contact info?	1	1589950823 NULL
11	11	3	0	6 Hey - nah I got it. You seen Gene ...	1	1589950899 NULL
12	12	4	0	7 I need to share my plan with Gene, ...	1	1589950899 NULL
13	13	4	0	7 Can't get ahold of Gene. Can you sen...	0	1589989999 NULL
14	14	4	0	7 SSB0YXZ1IGZvdW5kIGEGd2F5IHRvIGIha2Ug...	0	1589989999 NULL

Figure 9: DB Browser Browse Data tab of Zmsgs table highlighted is the encrypted message, its information, and full text.

Decode from Base64 format

Simply enter your data then push the decode button.

SSBoYXZlIGZvdW5kiGEgd2F5lHRvIG1ha2UgdGhllGRydWdzIHVuZGV0ZWNOYWJsZSBieSB0aGUgZG9ncyBh
dCB0aGUgY2hiY2twb2ludC4gSXQncyBteSBvd24gc3BIY2lhbCBtaXggb2Ygc2FnZWswcm9zZW1hc nkslHRoeW1l
LCB2YXBvcnVilGFuZCBjYXJvbGluYSByZWZwZXlqb2lsIQ==

 For encoded binaries (like images, documents, etc.) use the file upload form a little further down on this page.

UTF-8 Source character set.

☐ Decode each line separately (useful for when you have multiple entries).

☐ Live mode OFF Decodes in real-time as you type or paste (supports only the UTF-8 character set).

< DECODE > Decodes your data into the area below.

I have found a way to make the drugs undetectable by the dogs at the checkpoint. It's my own special mix of sage, rosemary, thyme, vaporub and carolina reaper oil!

Figure 10: Base64 Decoder with decoded text.

Additional Information

While going through the Zmsgs table, a photo was uncovered; its original data was converted to hexadecimal. The message itself is in Figure 11; the message was sent from Rett to OneZeroZeroPct@chattymail.com. Following this message was another message from OneZeroZeroPct@chattymail.com stating that it was “part of loot” they hit three months ago from that time.

The screenshot shows a database browser interface with the 'Zmsgs' table selected. The table has columns: zmsg_id, recip, auth, thread_id, text, read, date, and loc. Message 17 is highlighted in blue. The 'text' column for message 17 contains a long hexadecimal string. The 'read' column is set to 0, and the 'date' column is set to 1. The 'loc' column is set to 1.

zmsg_id	recip	auth	thread_id	text	read	date	loc
1	1	1	0	1 Hey Me Loeb, sorry I had to leave ...	1	1588436583	35°13'59.1"N 80°50'29.5"W
2	0	1	1	1 No issue Rett. We meet again next ...	1	1588436602	35°14'03.5"N 80°50'30.9"W
3	0	1	2	1 Hey Rett, I am missing a gold inlaid.	1	1588621803	35°14'03.5"N 80°50'30.9"W
4	0	5	3	1 When are you gonna get here? I'm ...	1	1588960203	35°14'49.4"N 80°48'20.5"W
5	5	0	3	1 Nu phone who die?	0	1588990423	N/A
6	0	2	4	1 Yo	1	1589030823	35°13'16.3"N 80°49'23.4"W
7	7	1	0	2 Hope. I haven't seen it.	1	1588629811	N/A
8	0	5	0	5 Oh, Rett., say mis yr mag. Been out ...	1	1589140823	N/A
9	0	1	2	2 Okay, let me know if you do find it.	1	1589950823	35°14'03.5"N 80°50'30.9"W
10	0	3	6	6 Cry. You need Gene's contact info?	1	1589950823	N/A
11	1	3	0	6 Hey - nah I got it. You seen Gene ...	1	1589950899	N/A
12	12	4	0	7 I need to share my plan with Gene, ...	1	1589950899	N/A
13	13	4	0	7 Can't get ahold of Gene. Can you see...	0	1589989999	N/A
14	14	4	0	7 888oYX310gWmK510gpd2F518Rv101ha20g...	0	1589989999	N/A
15	15	3	0	6 He said he "found" what I was lookin...	1	1589991234	N/A
16	16	3	0	6 Its this set of beauties	1	1589991235	N/A
17	17	3	0	6 FF D8 FF E0 00 10 4A 46 49 46 00 01 ...	1	1589991239	N/A
18	18	0	3	6 Hey, that's part of the loot from th...	0	1589991334	N/A
19	N/A	N/A	N/A	N/A	N/A	N/A	N/A
20	N/A	N/A	N/A	N/A	N/A	N/A	N/A
21	N/A	N/A	N/A	N/A	N/A	N/A	N/A

The hexadecimal data for message 17 is: FF D8 FF E0 00 10 4A 46 49 46 00 01 ...

Figure 11: Zmsgs table view of message 17 sent from Rett to OneZeroZeroPct@chattymail.com

DB Browser for SQLite - C:\Users\Sarah\Downloads\ChattyMcChatfaceChats.db

File Edit View Tools Help

New Database Open Database Write Changes Revert Changes Undo Open Project Save Project Attach Database Close Database

Database Structure Edit Pragas Execute SQL Browse Data

Table: Zmsgs

chat_ID	recip	auth	thread_ID	text	read	
Filter	Filter	Filter	Filter	Filter	Filter	Filter
6	6	0	2	4 Yo	1	1589
7	7	1	0	2 Nope. I haven't seen it.	1	1588
8	8	5	0	5 Oh, Rett.. sry mis yr msg. Been out ...	1	1589
9	9	0	1	2 Okay, let me know if you do find it.	1	1589
10	10	0	3	6 Crys. You need Gene's contact info?	1	1589
11	11	3	0	6 Hey - nah I got it. You seen Gene ...	1	1589
12	12	4	0	7 I need to share my plan with Gene, ...	1	1589
13	13	4	0	7 Can't get ahold of Gene. Can you sen...	0	1589
14	14	4	0	7 SSB0YXZ1IGZvdW5kIGEd2F5IHRvIGlha2Ug...	0	1589
15	15	3	0	6 He said he "found" what I was lookin...	1	1589
16	16	3	0	6 Its this set of beauties	1	1589
17	17	3	0	6 FF D8 FF E0 00 10 4A 46 49 46 00 01 ...	1	1589
18	18	0	3	6 Hey, that's part of the loot from th...	0	1589
19	19	NULL	NULL	NULL NULL	NULL	
20	20	NULL	NULL	NULL NULL	NULL	
21	21	NULL	NULL	NULL NULL	NULL	

5 - 21 of 21

Go to: 1

1 row(s), 7 column(s). Sum: 1.58999e+09; Average: 1.98749e+08; Min: 0; Max: 1.58999e+09

UTF-8

Edit Database Cell

Mode: RTL Text

Hey, that's part of the loot from the place we hit three months ago!

Editing row=18, column=5
Type: Text / Numeric; Size: 68 character(s)

Apply

DB Schema

Schema

```
CREATE TABLE "ZAllow" ( "Photos" INTEGER, "Contacts" INTEGER,
CREATE TABLE "Zmsgs" ( "Chat_ID" INTEGER, "recip" INTEGER, "z
CREATE TABLE "zConstants" ( "App_ID" INTEGER, "Ver" TEXT, "Ne
CREATE TABLE "zContacts" ( "Contact_ID" INTEGER, "Cnum" INTE
CREATE TABLE "zThread" ( "Thread_ID" INTEGER, "Start" INTEGE
```

Figure 12: Zmsgs table view of message 18 sent from OneZeroZeroPct@chattymail.com to Rett.

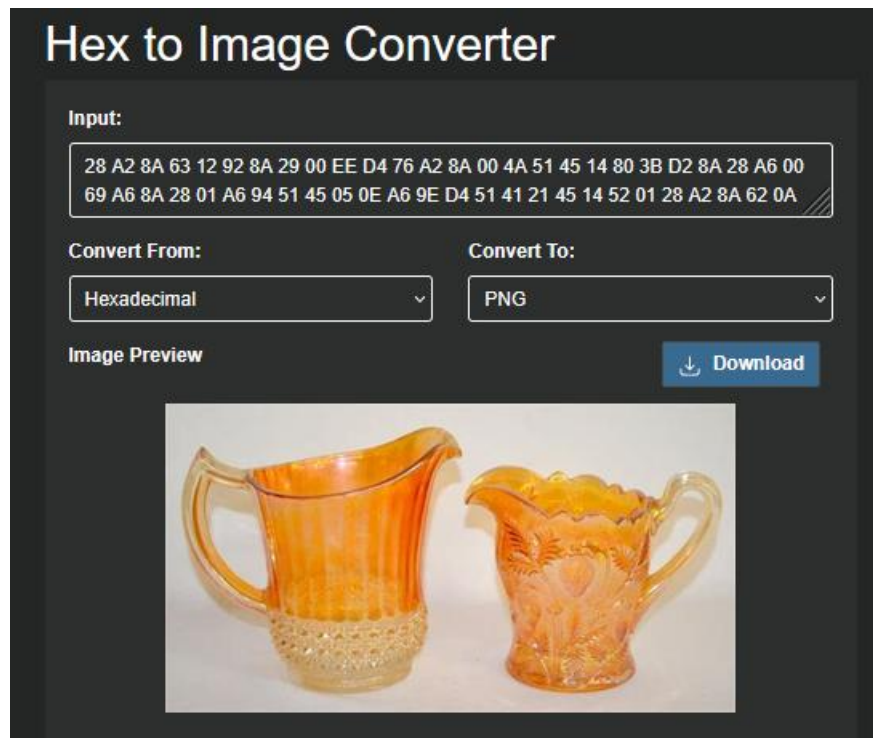


Figure 13: Hexadecimal to Image converter site showcasing the photo of items that were spoken about in message 18.

Conclusion

With this current analysis, Rett was talking to multiple people. Some of these people included his teacher, whom he might've stolen an item from. Another person was one of his possible accomplices, as they spoke about previous robberies; a message containing an encrypted message using Base64 with a very serious statement; another message containing a photo hidden within hexadecimal, followed by a message stating it was from a previous robbery they had committed. From this analysis, there are a lot of suspicious messages, with some containing hidden meanings and others being straight communication amongst some users.

Signed:

Sarah Lopez Martinez

Student Sarah Lopez Martinez

Student Forensics Examiner